

Denis Shcherba

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Education

Hermann-Ehlers-Schule , Highschool Diploma	Oct 2013 – June 2019
• GPA: 3.5/4.0	
Technical University of Berlin , BS in Computer Engineering	Oct 2019 – June 2023
• GPA: 2.4/4.0	
• Coursework: Digital Integrated Circuits, Scientific Computing, Embedded Programming, Hardware programming, FPGA programming, Circuit Design	
Technical University of Berlin , MS in Computer Engineering	Sept 2023 – Jan 2026
• estimated GPA: 3.5/4.0	
• Coursework: Robotics, Machine Learning, Robot Learning, Computer Vision, Robot Vision, Optimization, Differential Geometry	

Experience

Student Assistant , Learning and Intelligent Systems Group TU Berlin	May 2023 –
• Designed, built, and programmed an omnidirectional mobile base equipped with a robotic arm and multiple perception sensors	
• Developed a teleoperation system using motion capture for remote control of a dual-arm Franka Emika Panda setup	
• Benchmarked and fine-tuned various SLAM packages on a mobile manipulator to improve mapping and localization performance	
• Assisted new students in navigating the lab's robotics infrastructure and tools, providing hands-on support with their projects	
• Contributed to the robotics software stack, specifically the low-level control interface BotOp	
• Developed and maintained multiple robotic manipulation projects and supporting tools — see GitHub for details: Tupryk/SHK_LIS, denis-shcherba/LISLabWork, LIS-TU-Berlin	
Math Tutor	April 2020 -
• Provided personalized tutoring in advanced mathematics (multivariable calculus, linear algebra, differential equations) to over 15 students, improving their academic performance	

Publications

Meta-Optimization and Program Search using Language Models for Task and Motion Planning	May 2025
<i>Denis Shcherba*</i> , Eckart Cobo-Briesewitz*, Cornelius V. Braun, Marc Toussaint https://arxiv.org/abs/2505.03725 accpeted at Conference of Robot Learning 2025	

Extracurricular Activities

Vice President, FUSE RoboCup Rescue Team , TU Berlin — since 2024	Website
• Coordinated team logistics and programming efforts for autonomous robot development.	
• Responsible for advancements in SLAM, dynamics learning, inverse kinematics, and simulation.	
President, RAI Book Club , TU Berlin — since 2025	
• Organized and led discussions on diverse literary genres, enhancing critical thinking skills.	
• Fostered a vibrant community promoting interdisciplinary learning and intellectual exchange.	

Skills and Interests

Languages

German – Native

Russian – Native (active proficiency: B2)

English – Fluent

Programming

Languages: Python, C++, C, LaTeX

APIs & Tools: Linux, PyTorch, Git, ROS 2, IsaacLab, OpenCV, Open3D, robotic